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# Improving Attention-Enhanced Reservoir Computing with Intrinsic Plasticity

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*Dedicated to  
my family and my friends.*



## Abstract

While the Transformer architecture has dominated the machine learning field this decade, there is a growing search for more efficient architectures during both training and inference. Reservoir computing, while remaining a strong class of models for time series prediction, struggles with discrete data like text. Attention-enhanced reservoir computing (AERC) tries to overcome this limitation using attention only on a fixed size reservoir, avoiding a growing KV cache. In this paper, we integrate Intrinsic Plasticity to the fixed reservoir to improve the AERC architecture and test it on a wider set of tasks. We find TODO



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# Chapter 1

## Introduction

In recent years there has been a growing search for efficient, subquadratic alternatives to Transformer, to make training and inference cheaper. The main approaches are to use a lighter variation of attention, such as DeepSeek Sparse Attention, or the use a linear architecture, such as RWKV or DeltaNet. While Reservoir Computing is a class of architectures capable of state of the art prediction for time-series or speech recognition tasks, or more in general tasks that trains on continuous data, it struggles with discrete data, such as tokens.



## Chapter 2

# Literature Review

Text for the literature review.



## Chapter 3

# Methodology

Text for the methodology.



## Chapter 4

# Results

Text for the results.





## Chapter 5

# Discussion

Text for the discussion.



## Chapter 6

# Conclusion

Text for the conclusion.



# Bibliography